

Dirichlet-Multinomial Error-Choice Simulation

2026-09-28

1 Overview

This report evaluates **dirichlet-multinomial** sampling error across cell-type proportions generated from a weighted Beta(, 1) distribution.

1.1 Success rule

For each replicate and cell type j , the estimated proportions are averaged over the N people in that replicate. The average is compared with the population-level true proportion p_j :

$$\bar{p}_j = \frac{1}{N} \sum_{i=1}^N \hat{p}_{ij}, \quad AE_j = |\bar{p}_j - p_j|, \quad ARE_j = \frac{|\bar{p}_j - p_j|}{p_j}. \quad (1)$$

A replicate **succeeds** for AE if $\max AE_j \leq \tau_{AE}$, and for ARE if $\max ARE_j \leq \tau_{ARE}$. AE and ARE are always evaluated separately and never combined into one criterion.

2 Simulation setup

The data are generated in two steps (a Dirichlet-multinomial hierarchy):

1. Person i has a true composition $\theta_i \sim \text{Dirichlet}(c \cdot p)$, where p is the population mean composition and c is the **concentration**. The concentration is the sum of the parameters in a dirichlet-multinomial.
2. From that person, n cells are sampled: $\text{counts}_i \sim \text{Multinomial}(n, \theta_i)$, so $\hat{p}_{ij} = \text{count}_{ij}/n$.

The population proportions p are fixed values derived from a Beta distribution (Table 2). A larger α makes the composition more skewed, so the rarest types become rarer. Each scenario (α, N, n) is simulated B times. Table 1 lists all settings.

Table 1: Simulation settings (`simulation_dm_errorchoice_defaults()`).

Parameter	Value
alpha (Beta shape)	2, 3, 4, 5
K (cell types)	10
B (replicates per scenario)	1,000
Concentration c	10000
N (people per replicate)	1, 2, 3, 5, 10
n per person, sweep	100 to 100,000,000 (25 log-spaced values)
n per person, threshold sweep	200,000
tau_AE (fixed)	0.01
tau_ARE (fixed)	0.5
Target success rate	0.95
Seed	260926

Table 2: Population mean proportions p_j per cell type (columns) for each α (rows).

alpha	1	2	3	4	5	6	7	8	9	10
2	0.01	0.03	0.05	0.07	0.09	0.11	0.13	0.15	0.17	0.19
3	0.00075	0.0068	0.019	0.037	0.061	0.091	0.13	0.17	0.22	0.27
4	5e-05	0.0014	0.0063	0.017	0.037	0.067	0.11	0.17	0.25	0.34
5	3.2e-06	0.00026	0.002	0.0076	0.021	0.046	0.09	0.16	0.26	0.41

2.1 Concentration c

The **concentration** c controls the **biological variation** between people. The individual proportion θ_{ij} comes from a $Dirichlet(N, c \cdot p_j)$ such that:

$$E(\theta_{ij}) = p_j, \quad \text{Var}(\theta_{ij}) = \frac{p_j(1-p_j)}{c+1}. \quad (2)$$

So c acts as a precision parameter:

- **Small c :** people differ a lot. $\text{Var}(\theta_{ij})$ comes close to its maximum $p_j(1-p_j)$.
- **Large c :** everybody is close to p . As $c \rightarrow \infty$, the model reduces to a plain multinomial with no differences between people.

Conversely, if the between-person variance of a cell type is known from real data, $c = p_j(1-p_j) / \text{Var}(\theta_{ij}) - 1$.

Table 3 shows how large this variation is on the proportion scale, through the standard deviation $\text{SD}(\theta_{ij})$ and the coefficient of variation $\text{CV} = \text{SD}(\theta_{ij}) / p_j = \sqrt{(1-p_j)/(p_j(c+1))}$.

Table 3: Between-person variability of the individual proportions: $\text{SD}(\theta_{ij})$ with the coefficient of variation $\text{CV} = \text{SD} / p_j$ in brackets, for several concentrations c (columns) and population proportions p_j (rows).

	$c = 1,000$	$c = 10,000$	$c = 100,000$	$c = 1,000,000$
$p = 0.01$	0.0031 (0.31)	0.00099 (0.099)	0.00031 (0.031)	0.000099 (0.0099)
$p = 0.05$	0.0069 (0.14)	0.0022 (0.044)	0.00069 (0.014)	0.00022 (0.0044)
$p = 0.1$	0.0095 (0.095)	0.003 (0.03)	0.00095 (0.0095)	0.0003 (0.003)
$p = 0.4$	0.015 (0.039)	0.0049 (0.012)	0.0015 (0.0039)	0.00049 (0.0012)

At the simulated concentration $c = 10,000$, a cell type with a population mean of 10% has a between-person SD of 0.003, so 95% of people have between 9.4% and 10.6% of that type. For rare types the

relative spread is larger. For $p_j = 0.01$ (CV 0.099), 95% of people have between 0.81% and 1.20%. Only for the very rarest types, where $c p_j \ll 1$ (down to $c p_j \approx 0.032$ at $\alpha = 5$), is the distribution of θ_{ij} heavily skewed. There, almost everyone has essentially none of that cell type, and a few people have a lot.

3 Simulation results

3.1 Success rate versus threshold

Figure 1 and Figure 2 show the success rate as a function of the threshold τ at a fixed 200,000 cells per person.

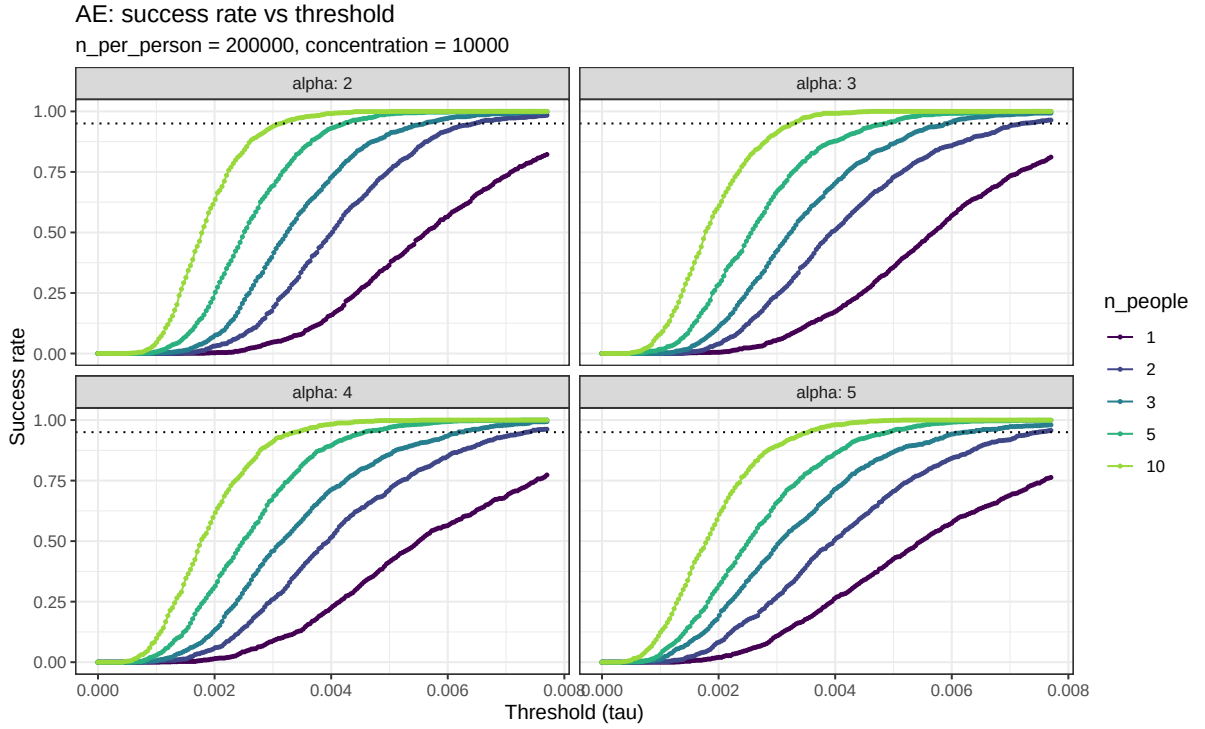


Figure 1: AE success rate versus threshold τ_{AE} at 200,000 cells per person. One panel per α , one colour per number of people N ; the dotted line marks the 95% target.

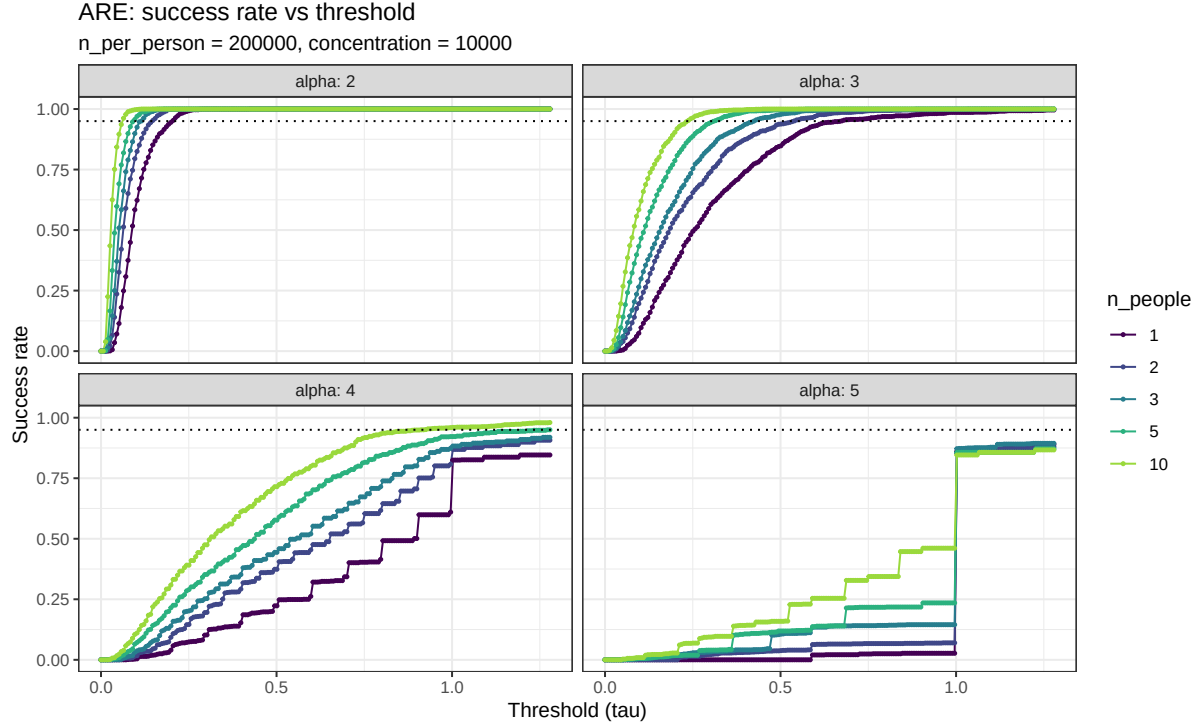


Figure 2: ARE success rate versus threshold τ_{ARE} at 200,000 cells per person. Note the jump at $\tau_{ARE} = 1$.

3.2 Success rate versus cells per person

Figure 3 and Figure 4 fix the threshold ($\tau_{AE} = 0.01$, $\tau_{ARE} = 0.5$) and vary the number of cells per person.

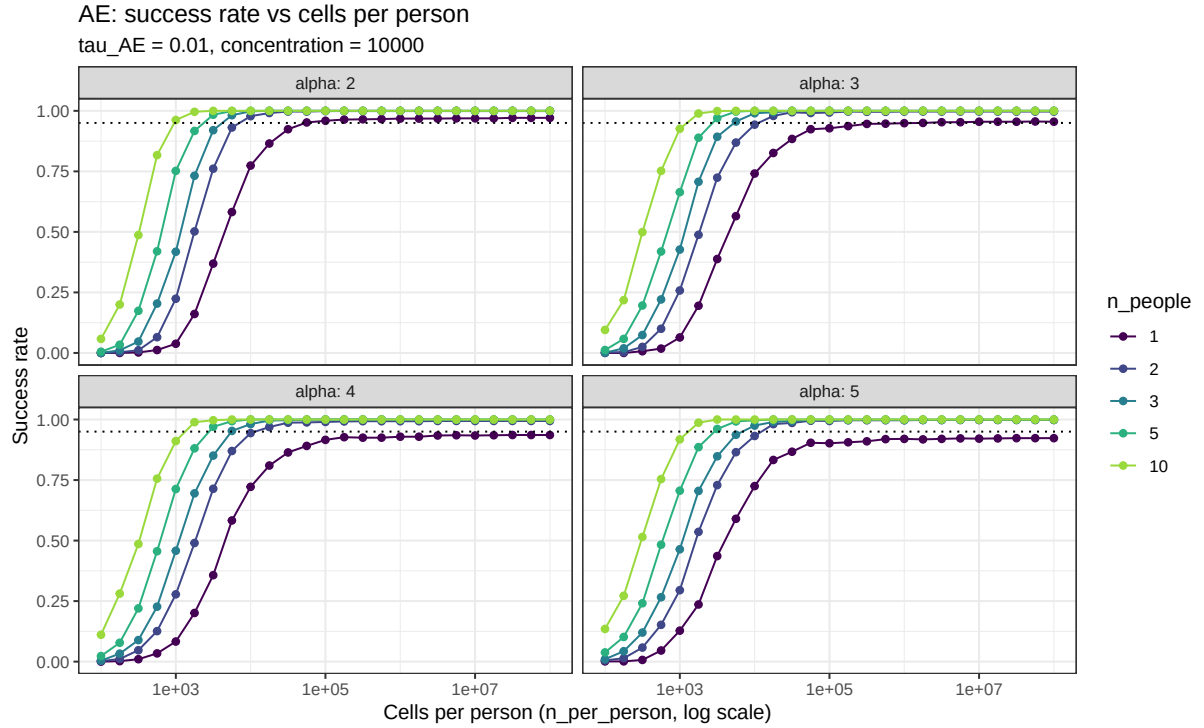


Figure 3: AE success rate versus cells per person (log scale) at $\tau_{AE} = 0.01$. Success rises with cells per person and levels off from roughly $n \approx c$ onwards.

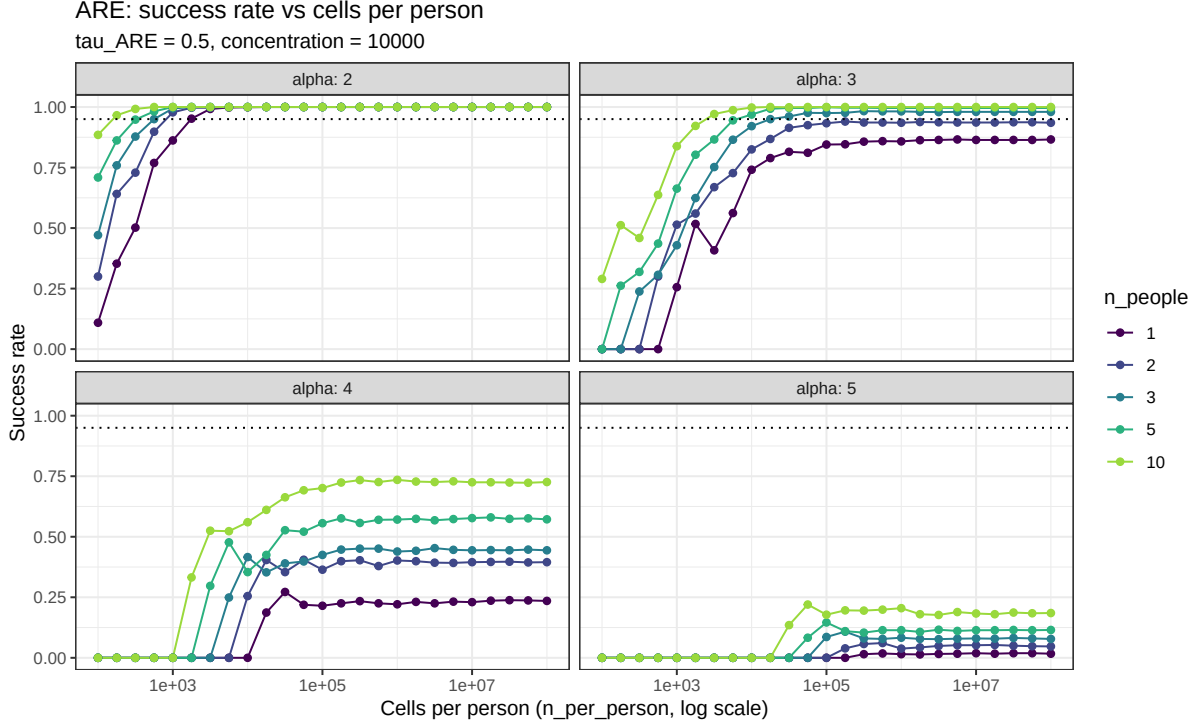


Figure 4: ARE success rate versus cells per person (log scale) at $\tau_{ARE} = 0.5$.

For AE, the success rate increases with the number of cells per person, but only up to a point. At $\alpha = 5$ with a single person, it rises from 0% at 10^2 cells to 72% at $n \approx c$, and ends at 92% at 10^8 cells. With 10 people the same curve ends at 100%. Beyond roughly 10^5 cells per person, extra cells hardly help, and only more people raise the plateau (explanation: see Section 4).

For ARE a similar result can be seen, success rate increases with the number of cells per person, but only up to a point, after which only an increase in the number of people can raise the success rate. Due to the rarity of the rarest cell types, at $\alpha = 5$ it ends at only 2% (1 person) and 18% (10 people). The rarest cell types have $cp_j \ll 1$, so in most replicates none of the people carry them. Then $\bar{p}_j = 0$ and $ARE_j = 1 > \tau_{ARE}$, however many cells are counted.

3.3 Required cells per person

Figure 5 and Figure 6 show, for each α and number of people N , the smallest number of cells per person for which the success rate reaches the 95% target at the fixed thresholds, found with an iterative sample-size solver. Before solving, it is checked whether the target can be reached at all with very many cells per person, or if the success rate plateaus under the wanted success rate. Those combinations where it cannot are marked with $\times$ at the top of the panel. Open markers indicate that the algorithm did not converge and indicate the last result before converging.

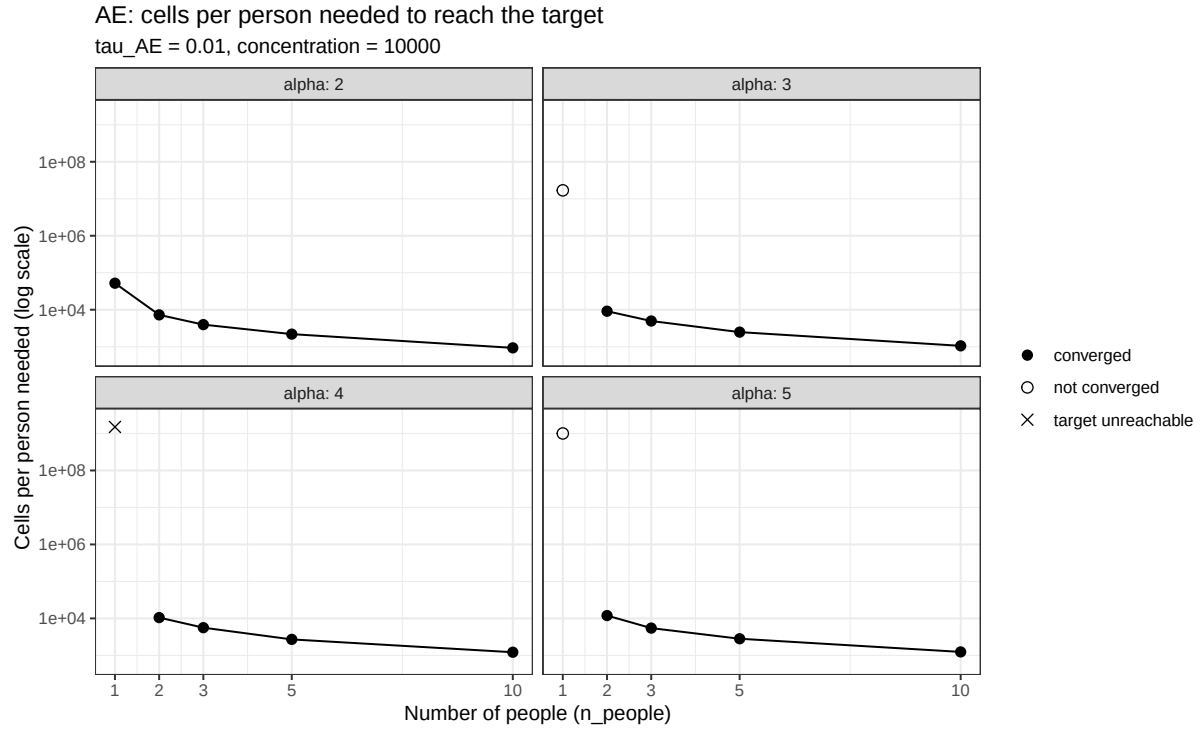


Figure 5: Cells per person needed to reach the 95% AE success target at $\tau_{AE} = 0.01$ (log scale), versus the number of people N . One panel per α . An open marker means the solver did not converge; $\times$ at the top means the target cannot be reached.

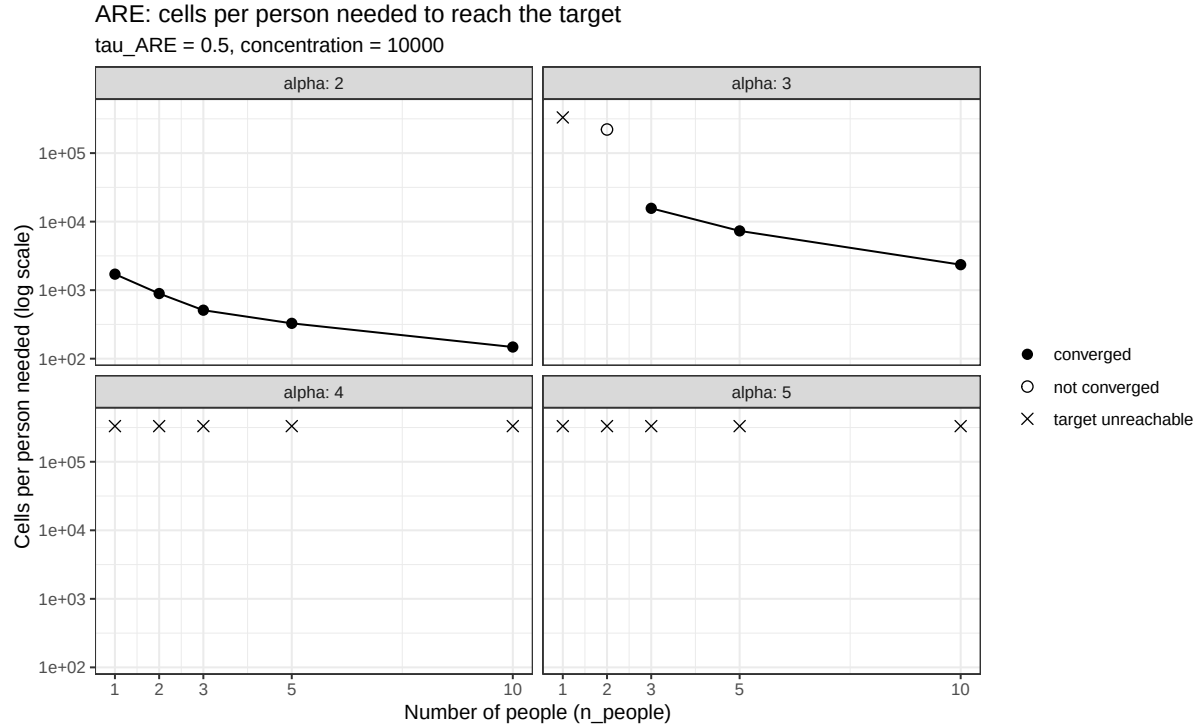


Figure 6: Cells per person needed to reach the 95% ARE success target at $\tau_{ARE} = 0.5$ (log scale), versus the number of people N . One panel per α . An open marker means the solver did not converge; $\times$ at the top means the target cannot be reached.

4 Why the curves level off

4.1 Error sources

Given the two-step generation in Section 2, the error of $\bar{p}_j$ has two parts:

- **Differences between people (biological variation).** Each θ_{ij} scatters around p_j with variance $p_j(1 - p_j)/(c + 1)$ (Equation 2).
- **Sampling (technical variation).** Given the person, each $\hat{p}_{ij}$ scatters around θ_{ij} with variance $\theta_{ij}(1 - \theta_{ij})/n$. On average over people, this is $E[\theta_{ij}(1 - \theta_{ij})]/n = \frac{p_j(1 - p_j)c}{(c + 1)n}$.

Assuming people are independent, averaging over N of them gives

$$\text{Var}(\bar{p}_j - p_j) = p_j(1 - p_j) \left[\underbrace{\frac{1}{N(c + 1)}}_{\text{biological}} + \underbrace{\frac{c}{(c + 1)Nn}}_{\text{technical}} \right]. \quad (3)$$

- The **biological term** depends only on the number of people N and the concentration c .
- The **technical term** depends only on the **total** number of cells counted, $N \cdot n$.

The ratio of technical to biological variance is c/n . So technical variation makes up a share $c/(c + n)$ of the variance:

- For $n \ll c$, technical variation dominates, and more cells per person clearly improve the estimate. This is the rising part of Figure 3 and Figure 4.
- At $n = c = 10,000$ the two parts are equal.
- At $n = 10^5$ the technical share is only 9%. From there on, the SD of $\bar{p}_j$ is within 5% of its floor, and the curves level off.
- Even with infinitely many cells per person, $\bar{p}_j$ converges to the average composition of *those particular* N people, and not to p_j . The biological term stays, and only more people lower the plateau.

4.2 Illustration with Additional Simulation

To illustrate Equation 3 directly per cell type, I have created a small additional simulation. For $\alpha = 5$ it draws θ_{ij} and $\text{count}_{ij} \sim \text{Binomial}(n, \theta_{ij})$. This is done for the most abundant cell type (cell type 10) and for a rare one (cell type 2).

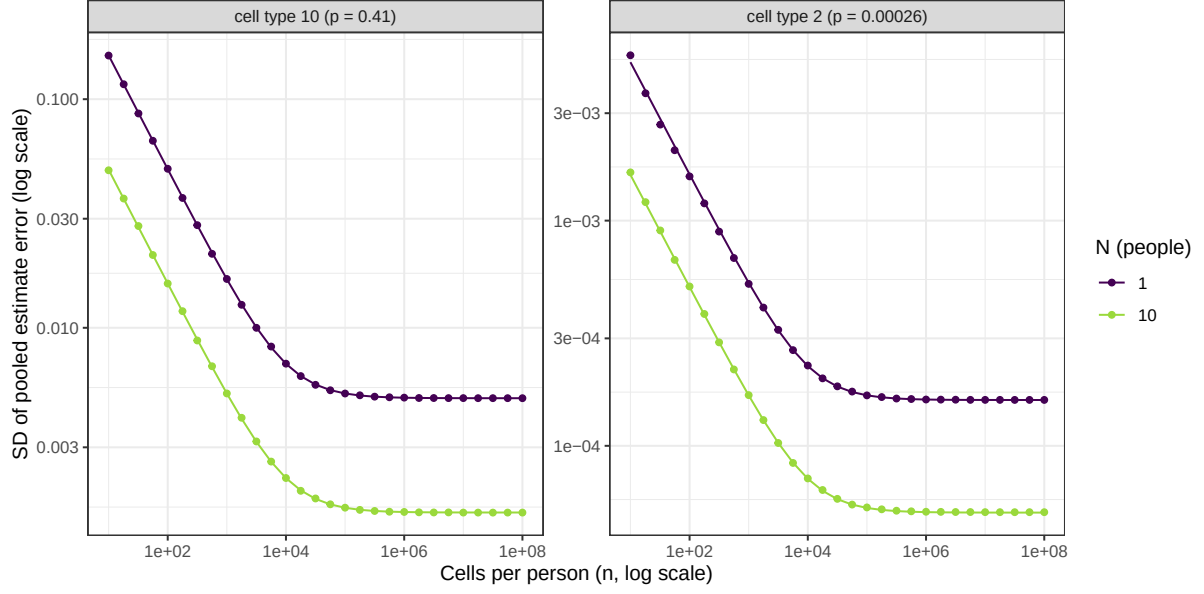


Figure 7: Standard deviation of $\bar{p}_j - p_j$ versus cells per person (both axes log scale) at $\alpha = 5$, $c = 10,000$. Points: simulated (20,000 replicates). Lines: Equation 3.

Figure 7 illustrates the decomposition Equation 3. The SD falls with n up to about $n \approx c$ and then flattens, and above roughly 10^5 cells adding cells no longer helps.